Integrating Other Functions

Exam code: 9MA0

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Written by: Paul
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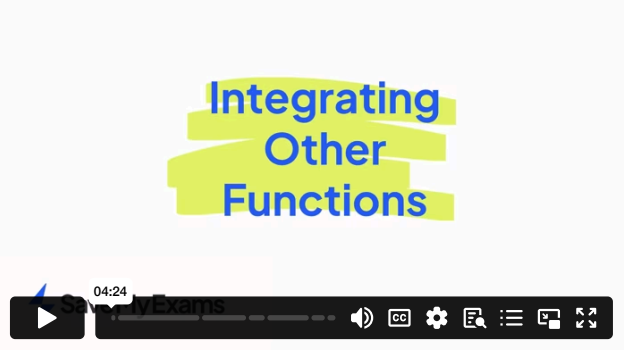
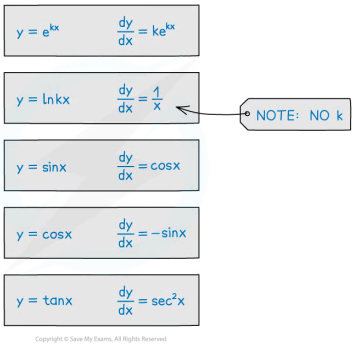


Did this video help you? Yes No

Integrating other functions

Is integration the reverse of differentiation?

- Yes, but remember "+c", the constant of integration ...
  - ... unless finding a definite integral
- Recognising common results helps to make integration easier (See Differentiating Other Functions)



How do I integrate e^x and e^{kx}?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

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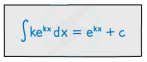
- The gradient of e^{kx} is ke^{kx}
  - ie $y = e^{kx}$, $dy/dx = ke^{kx}$
- The reverse applies when integrating
- This is an example of **reverse chain rule**



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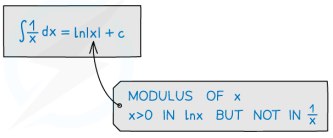
How do I integrate 1/x?



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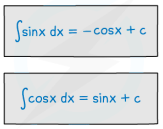
- Remember $\frac{1}{x} = x^{-1}$
  - The method for integrating powers does **not** apply if the power is -1

How do I integrate sin x and cos x?



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- Note the minus in the integral of **sin x**

How do I integrate tan x?

- The integral of **tan x** is **ln|sec x| + c**

Examiner Tips and Tricks

- Make sure you have a copy of the formula booklet during revision but don't try to remember everything in the formula booklet.
- However, do be familiar with the **layout** of the formula booklet – you'll be able to locate quickly whatever you are after, and you do not want to be searching every line of every page!

Worked Example

Find the following integrals

(a) $\int 3e^{3x}\,dx$

(b) $\int_{-6}^{-3} \frac{1}{x}\,dx$

(c) $\int \sin x\,dx$

a) $\int 3e^{3x}dx = e^{3x} + c$

IN THE FORM ke^{kx}

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